

Multi-Agent Scientific Paper Review

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File	buildings-15-03743 (3).pdf
Words	12,561
Sections Detected	preamble, introduction, section_1, abstract, references, title, keywords
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MAJOR REVISION

Weighted Score: 2.8 / 5.0
Confidence: 80%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	3.0	Acceptable	■■■■■
Manuscript Structure	1.0	Very Weak	■■■■■
Methodology & Equations	3.5	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	3.0	Acceptable	■■■■■
Discussion & Conclusions	1.5	Very Weak	■■■■■
Figures & Tables	3.5	Good	■■■■■
Scientific Writing	2.5	Weak	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	1.5	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 3.0 / 5.0 | Confidence: 90%

Strengths:

- [TITLE] The title attempts to reflect the use of classical and quantum-inspired methods in parameter identification.
- [ABSTRACT] The abstract clearly states the research problem and quantifies results with specific percentages.
- [KEYWORDS] The keywords include domain-specific terms such as 'seismic engineering' and 'multi-objective optimization'.

Weaknesses:

- [TITLE] The title is incomplete and lacks clarity, making it difficult to determine the study's focus.
- [ABSTRACT] The abstract is incomplete and does not provide a standalone summary of the study.
- [KEYWORDS] The keywords are not fully distinct from the title and lack some domain-specific indexing terms.

Major Comments:

- [TITLE] The title is incomplete and does not clearly reflect the study's contribution. It should be revised to include the focus on seismic engineering and parameter identification.
- [ABSTRACT] The abstract is incomplete and ends abruptly, making it difficult to assess the full scope of the study. It should be revised to ensure it is standalone and complete.

Minor Comments:

- [KEYWORDS] The keywords should be revised to avoid duplication with the title and include more domain-specific indexing terms.

Recommendations:

- [TITLE] Revise the title to be more specific and complete, reflecting the focus on seismic engineering and parameter identification.
- [ABSTRACT] Revise the abstract to ensure it is complete and follows a structured format with clear sections for background, objective, methods, results, and conclusions.
- [KEYWORDS] Revise the keywords to avoid duplication with the title and include more domain-specific indexing terms.

StructureReviewer

Manuscript structure and logical coherence

Score: 1.0 / 5.0 | Confidence: 90%

Strengths:

- Clear title and keywords
- Abstract outlines the study's purpose

Weaknesses:

- Missing critical sections (literature review, methods, results, discussion, conclusions)
- Logical flow is severely disrupted
- Research problem and hypothesis are not clearly stated

Major Comments:

- The manuscript lacks essential sections such as literature review, methods, results, discussion, and conclusions, which are fundamental for a scientific paper.
- The logical flow is not present, and the manuscript does not follow the standard structure of Introduction → Literature → Methods → Results → Discussion → Conclusions.

Minor Comments:

- The abstract is incomplete and does not clearly state the research problem or hypothesis.

- The title is not properly formatted and lacks clarity.

Recommendations:

- The authors are strongly advised to restructure the manuscript to include all required sections (literature review, methods, results, discussion, and conclusions).
- The abstract should be revised to clearly state the research problem, hypothesis, and objectives.
- The title should be properly formatted and clearly reflect the study's focus.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear problem definition and motivation
- Logical flow from problem to model to method to experiments
- Use of well-known algorithms with quantum-inspired enhancements
- Use of a five-story building model for testing
- Comparison with multiple baselines (GA, PSO, etc.)

Weaknesses:

- Lack of detailed description of the exact implementation of quantum-inspired algorithms
- Insufficient detail on the exact experimental setup and parameters used
- No mention of the exact metrics used for performance evaluation
- No information on the computational resources used

Major Comments:

- The manuscript lacks sufficient detail on the exact implementation of quantum-inspired algorithms, which is critical for reproducibility and understanding of the method.

Minor Comments:

- The experimental setup and parameters used in the five-story building model should be described in more detail for reproducibility.
- The performance metrics used for evaluating the algorithms are not clearly defined.

Recommendations:

- Provide a detailed description of the implementation of quantum-inspired algorithms, including the quantum rotation matrix and its application in the optimization process.
- Include a detailed description of the experimental setup, including the exact parameters of the five-story building model and the seismic excitation conditions.
- Clarify the performance metrics used to evaluate the algorithms, such as convergence speed, accuracy, and robustness to noise.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and Bootstrap confidence intervals for non-normal data
- Reporting of RMSE, MSE, and CV for performance evaluation
- Use of multiple comparison corrections (Tukey-Bonferroni)
- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Weaknesses:

- Lack of effect size reporting
- No distinction between statistical and practical significance
- No reproducibility details
- No justification for 30 independent runs

Major Comments:

- The use of ANOVA without verification of normality and homoscedasticity is inappropriate. The manuscript should use nonparametric methods or justify the use of parametric tests.
- The manuscript fails to report effect sizes, which are critical for interpreting the magnitude of differences between methods.
- There is no distinction between statistical and practical significance, which limits the interpretation of results.
- The reproducibility of results is not addressed, which is essential for scientific rigor.

Minor Comments:

- The manuscript should clarify the justification for using 30 independent runs.
- Bootstrap confidence intervals should specify the number of resamples used.
- The coefficient of variation (CV) should be interpreted in the context of the data.

Recommendations:

- Report effect sizes (e.g., Cohen's d, eta-squared) to complement p-values.
- Clarify the distinction between statistical and practical significance.
- Provide details on reproducibility, such as random seeds and variance in results.
- Specify the number of bootstrap resamples used to construct confidence intervals.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 62%

Strengths:

- 21 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.

ResultsReviewer

Results validity and empirical support

Score: **3.0 / 5.0** | Confidence: 70%

Strengths:

- MSE, RMSE, CV, and hypervolume metrics are reported, indicating quantitative analysis
- Baseline comparisons with GA and PSO are included
- Pareto analysis and spread indicators are mentioned, suggesting multi-objective optimization evaluation

Weaknesses:

- The results section is cut off before completion, making it impossible to assess full validity and completeness
- No mention of limitations or negative results, which is critical for transparency
- No clear statement about how results are consistent across text, figures, and tables

Major Comments:

- The results section is incomplete and cut off, preventing a full evaluation of validity and completeness.

Minor Comments:

- The lack of explicit mention of limitations and negative results reduces the transparency of the findings.

Recommendations:

- Complete the results section to provide full empirical support and clarity.
 - Include a discussion of limitations and negative results to enhance transparency.
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **1.5 / 5.0** | Confidence: 90%

Strengths:

- References to relevant literature
- Potential for further exploration

Weaknesses:

- Lack of discussion on limitations
- No direct derivation of conclusions from results
- No clear comparison with prior work
- Overgeneralization of findings
- No specific future work directions

Major Comments:

- The Discussion and Conclusions sections are insufficiently developed. They lack critical analysis, comparison with prior work, and clear limitations. Conclusions are not directly derived from results and fail to address the original hypothesis. Overgeneralization is evident, and future work is not specified or justified.

Minor Comments:

- The manuscript lacks a clear discussion of the study's limitations and theoretical implications. The conclusions are vague and not supported by the data presented.

Recommendations:

- Revise the Discussion section to include a thorough comparison with prior work and address any unexpected findings. Clearly state the limitations of the study. Ensure that the Conclusions are directly derived from the results and answer the original hypothesis. Avoid overgeneralization and specify concrete directions for future work.
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical terminology in equations
- Use of specific acronyms with definitions in context

Weaknesses:

- Overuse of passive voice
- Unexplained acronyms
- Vague quantifiers
- Incomplete sentences
- Poor paragraph structure

Major Comments:

- The manuscript contains several instances of unclear writing, including unexplained acronyms and vague quantifiers. The use of passive voice is excessive and hinders clarity. Several sentences are incomplete and require rewriting for coherence and completeness.

Minor Comments:

- Some equations are not properly formatted or labeled
- Some acronyms are introduced without explanation

Recommendations:

- Define all acronyms on first use
- Rewrite sentences with passive voice to active voice
- Complete all sentences and ensure logical flow between paragraphs
- Clarify vague quantifiers with specific numbers or examples

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Citations are consistent in format
- Recent references (2021-2025) are included

Weaknesses:

- Lack of seminal works in quantum computing or optimization
- Missing key areas in quantum-inspired algorithms
- Irrelevant references to structural engineering

Major Comments:

- The references section lacks seminal works in quantum computing and optimization, which are critical for the manuscript's topic.
- The inclusion of structural engineering references is irrelevant to the quantum-inspired optimization focus of the manuscript.

Minor Comments:

- The citation style is consistent, but the references section could be more focused on the quantum computing and optimization domain.

Recommendations:

- Include seminal works in quantum computing and optimization algorithms.
- Remove irrelevant references to structural engineering.
- Ensure the references section is focused on the quantum-inspired optimization domain.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- The manuscript explicitly states limitations, although they are not sufficiently detailed or specific.
- The study presents a novel approach combining classical and quantum-inspired methods for parameter identification, which is a significant contribution.

Weaknesses:

- The limitations section lacks specificity and depth, which undermines the credibility of the study's conclusions.
- The manuscript does not address reproducibility, which is critical for scientific validation.

- The absence of ethical approval, data availability, conflict of interest, and funding disclosure raises serious concerns about transparency and compliance with COPE standards.

Major Comments:

- The manuscript fails to report essential ethical and transparency information, which is required for publication in Q1 journals.
- The limitations section is not sufficiently detailed or specific, which weakens the study's credibility and impact.

Minor Comments:

- The study presents a novel approach, which is a strength.

Recommendations:

- The authors are strongly advised to include ethical approval details, data availability statements, conflict of interest declarations, and funding disclosures.
- The limitations section should be expanded to provide more specific and honest details about the study's constraints and their impact on the conclusions.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [TITLE] The title is incomplete and does not clearly reflect the study's contribution. It should be revised to include the focus on seismic engineering and parameter identification.
- [2] [ABSTRACT] The abstract is incomplete and ends abruptly, making it difficult to assess the full scope of the study. It should be revised to ensure it is standalone and complete.
- [3] The manuscript lacks essential sections such as literature review, methods, results, discussion, and conclusions, which are fundamental for a scientific paper.
- [4] The logical flow is not present, and the manuscript does not follow the standard structure of Introduction → Literature → Methods → Results → Discussion → Conclusions.
- [5] The manuscript lacks sufficient detail on the exact implementation of quantum-inspired algorithms, which is critical for reproducibility and understanding of the method.
- [6] The use of ANOVA without verification of normality and homoscedasticity is inappropriate. The manuscript should use nonparametric methods or justify the use of parametric tests.
- [7] The manuscript fails to report effect sizes, which are critical for interpreting the magnitude of differences between methods.
- [8] There is no distinction between statistical and practical significance, which limits the interpretation of results.
- [9] The reproducibility of results is not addressed, which is essential for scientific rigor.
- [10] The results section is incomplete and cut off, preventing a full evaluation of validity and completeness.
- [11] The Discussion and Conclusions sections are insufficiently developed. They lack critical analysis, comparison with prior work, and clear limitations. Conclusions are not directly derived from results and fail to address the original hypothesis. Overgeneralization is evident, and future work is not specified or justified.
- [12] The manuscript contains several instances of unclear writing, including unexplained acronyms and vague quantifiers. The use of passive voice is excessive and hinders clarity. Several sentences are incomplete and require rewriting for coherence and completeness.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [KEYWORDS] The keywords should be revised to avoid duplication with the title and include more domain-specific indexing terms.
- [2] The abstract is incomplete and does not clearly state the research problem or hypothesis.
- [3] The title is not properly formatted and lacks clarity.
- [4] The experimental setup and parameters used in the five-story building model should be described in more detail for reproducibility.
- [5] The performance metrics used for evaluating the algorithms are not clearly defined.
- [6] The manuscript should clarify the justification for using 30 independent runs.
- [7] Bootstrap confidence intervals should specify the number of resamples used.
- [8] The coefficient of variation (CV) should be interpreted in the context of the data.
- [9] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.
- [10] The lack of explicit mention of limitations and negative results reduces the transparency of the findings.
- [11] The manuscript lacks a clear discussion of the study's limitations and theoretical implications. The conclusions are vague and not supported by the data presented.
- [12] Some equations are not properly formatted or labeled

Specific Improvement Recommendations

1. [TITLE] Revise the title to be more specific and complete, reflecting the focus on seismic engineering and parameter identification.
2. [ABSTRACT] Revise the abstract to ensure it is complete and follows a structured format with clear sections for background, objective, methods, results, and conclusions.
3. [KEYWORDS] Revise the keywords to avoid duplication with the title and include more domain-specific indexing terms.
4. The authors are strongly advised to restructure the manuscript to include all required sections (literature review, methods, results, discussion, and conclusions).
5. The abstract should be revised to clearly state the research problem, hypothesis, and objectives.
6. The title should be properly formatted and clearly reflect the study's focus.
7. Provide a detailed description of the implementation of quantum-inspired algorithms, including the quantum rotation matrix and its application in the optimization process.
8. Include a detailed description of the experimental setup, including the exact parameters of the five-story building model and the seismic excitation conditions.
9. Clarify the performance metrics used to evaluate the algorithms, such as convergence speed, accuracy, and robustness to noise.
10. Report effect sizes (e.g., Cohen's d, eta-squared) to complement p-values.
11. Clarify the distinction between statistical and practical significance.
12. Provide details on reproducibility, such as random seeds and variance in results.